

Q1. What are Power Pivot components and what are its tasks?

Power Pivot is a data modeling and analysis technology from Microsoft. It allows users to work with large amounts of data, create relationships between tables, and perform calculations using DAX (Data Analysis Expressions).

Main components of Power Pivot

1. Data Model

Stores multiple tables together and allows relationships to be created between them.

2. Tables

Contain data organized into rows and columns. For example, Sales, Product, Customer, and Date tables.

3. Relationships

Connect tables using common columns such as Product ID, Customer ID, or Date ID.

4. DAX

Used to create calculations such as measures and calculated columns.

5. Power Pivot Window / Data View

Used to view, manage, and work with the data model and tables.

Main tasks of Power Pivot

- Import data from different sources.
- Combine multiple tables.
- Create relationships between tables.
- Create calculated columns.
- Create measures using DAX.
- Analyze large datasets.
- Build data models for reporting.
- Perform calculations and generate business insights.

In simple words:

Power Pivot → Import Data → Create Relationships → Create DAX
Calculations → Analyze Data

Q2. What are the different views in Power BI Desktop? Explain each and its working.

Power BI Desktop mainly provides three important views for working with data.

1. Report View

Purpose: Used to create interactive reports and visualizations.

In Report View, we can create:

- Bar charts
- Column charts
- Line charts
- Pie charts
- Tables
- Cards
- Maps
- Slicers

Working:

Select Data → Choose Visual → Add Fields → Format Visual → Create Report

2. Table/Data View

Purpose: Used to view and inspect the data loaded into Power BI.

It displays the rows and columns of tables.

We can also:

- Check data values.
- Check columns.
- Create calculated columns.
- Create measures.
- Examine the data before creating reports.

Working:

Load Data → Open Data View → Inspect Tables → Create Calculations

3. Model View

Purpose: Used to create and manage relationships between tables and understand the data model.

It displays:

- Tables
- Columns
- Primary/unique keys
- Relationships
- Cardinality

Working:

Tables → Identify Common Keys → Create Relationships → Build Data Model

Q3. What are Star and Snowflake Schema? Explain with diagrams.

A. Star Schema

A Star Schema is a data-modeling structure in which a central Fact Table is directly connected to multiple Dimension Tables.

It is called a **Star Schema** because the structure looks like a star.

Characteristics of Star Schema

- One central fact table.
- Multiple dimension tables.
- Dimensions are generally denormalized.
- Simple structure.
- Easy to understand.

- Commonly used in Power BI and Business Intelligence.
- Generally provides efficient analytical queries.

B. Snowflake Schema

A Snowflake Schema is an extension of the Star Schema where dimension tables are normalized into multiple related tables.

Characteristics of Snowflake Schema

- Contains a central fact table.
- Dimension tables can be divided into multiple related tables.
- Dimensions are normalized.
- Reduces data duplication.
- More complex than a Star Schema.
- Has more relationships to manage.

Q4. What is a Dimension and Fact Table?

Fact Table

A Fact Table contains the measurable business transactions or numerical information used for analysis.

For example, a Sales fact table may contain:

- Sales Amount
- Sales Quantity
- Discount Amount
- Cost
- Product Key
- Date Key
- Store Key
- Customer Key

Example:

FACT SALES

Sales ID

Date Key

Product Key

Store Key

Quantity

Sales Amount

Discount

Profit

The fact table is normally located at the center of a Star or Snowflake Schema.

Dimension Table

A Dimension Table contains descriptive information that provides context for analyzing the facts.

Examples include:

- Date
- Product
- Customer
- Store
- Geography
- Channel

Example:

DIM PRODUCT

Product Key

Product Name

Category

Subcategory

Brand

Color

For example, the Sales table tells us how much was sold, while the Product dimension tells us what product was sold.

